

```

In[1]:= If[! TrueQ[Private`$ev], Block[{Private`$ev = True},
  Print["Total time: ", First@AbsoluteTiming[ClearAll["Global`*"];
    NotebookEvaluate[EvaluationNotebook[], InsertResults -> True]], " s"]]]

(* =====*)
(*Verification of low Mach number expansion for u and φ up to 8th order*)
(*All equations should yield {0,0} or 0. If all pass,prints True.*)
(* =====*)

(*_ 0th order _*)
u0 = {(3 Tf) / (4 (3 μ + α)) (2 μ + α - α x2^2), 0};

(*_ 2nd order _*)
φ2 = (3 Tf (2 - x1) α μ) / (2 (α + 3 μ));

u2 = {1 / (320 (α + 3 μ)^3) × 3 Tf^2 α (80 μ (α + 3 μ) ((-3 + x1) α + 3 (-2 + x1) μ) +
  Tf x2^2 α (2 (7 - 5 x2^2 + x2^4) α - 15 (-2 + x2^2) μ)),
  - ((3 Tf^2 x2 (-1 + x2^2) α^2 μ) / (8 (α + 3 μ)^2))};

(*_ 4th order _*)
φ4 = - (1 / (40 (α + 3 μ)^3)) 3 Tf^2 α μ (Tf x1 (4 α^2 + 30 α μ + 45 μ^2) +
  15 (α + 3 μ) ((-1 + 2 x2^2) α μ + (-1 + x2^2) α ν - 2 μ (μ + ν)));

u4 = {- (1 / (716800 (α + 3 μ)^5)) 3 Tf^3 α
  (-480 Tf μ (α + 3 μ) ((79 x1 + 70 (-3 + x1) x2^2 - 35 (-3 + x1) x2^4) α^3 + 7 (112 x1 +
    30 (-2 + x1) x2^2 - 15 (-2 + x1) x2^4) α^2 μ + 2520 x1 α μ^2 + 2520 x1 μ^3) +
  Tf^2 x2^2 α ((-1836 - 1670 x2^2 + 3052 x2^4 - 1155 x2^6 + 128 x2^8) α^3 +
    15 (-2360 + 1036 x2^2 + 252 x2^4 - 117 x2^6) α^2 μ -
    5040 (27 - 15 x2^2 + x2^4) α μ^2 + 75600 (-2 + x2^2) μ^3) +
  33600 μ (α + 3 μ)^2 (μ ((-42 - 8 (-5 + x1) x1 - 2 x2^2 + x2^4) α^2 -
    24 (-3 + x1) (-1 + x1) α μ + 24 μ^2) + 6 (α + 2 μ)^2 ν)),
  1 / (4480 (α + 3 μ)^4) × 9 Tf^3 x2 (-1 + x2^2) α^2 μ (-280 (-2 + x1) α μ (α + 3 μ) +
    Tf ((3 + x2^2) (-11 + 2 x2^2) α^2 - 280 α μ - 420 μ^2))};

(*_ 6th order _*)
φ6 = - (1 / (22400 (α + 3 μ)^5)) 9 Tf^3 α μ
  (Tf^2 x1 (258 α^4 + 2795 α^3 μ + 12495 α^2 μ^2 + 25200 α μ^3 + 18900 μ^4) +
    1400 α μ (α + 3 μ)^2 (2 (6 - 5 x1 + 3 (-2 + x1) x2^2) α μ +
      (12 - 5 x1 + 3 (-2 + x1) x2^2) α ν - 12 μ (2 (-1 + x1) μ + (-2 + x1) ν)) + 35 Tf (α + 3 μ)
    (-360 μ^3 (μ + ν) + 60 α μ^2 ((-7 + 6 (-4 + x1) x1 + 6 x2^2) μ + (-7 + 3 x2^2) ν) +
      2 α^3 ((-8 + 20 (-6 + x1) x1 + 17 x2^2 - 5 x2^4 + 3 x2^6) μ +
        (-8 + x2^2 + 5 x2^4 - x2^6) ν) +
      α^2 μ (8 (-19 + 30 (-5 + x1) x1 + 30 x2^2) μ + (-152 + 15 x2^2 (6 + x2^2)) ν)));

u6 = {- (1 / (114802688000 (α + 3 μ)^7)) 3 Tf^4 α
  (Tf^3 x2^2 α ((15274892 - 129985310 x2^2 + 94610516 x2^4 + 15506205 x2^6 -
    28058030 x2^8 + 7152145 x2^10 - 540480 x2^12) α^5 + 1365
    (148856 - 1403248 x2^2 + 1297296 x2^4 - 293436 x2^6 - 21714 x2^8 + 8011 x2^10)
    α^4 μ + 720720 (-2901 - 8155 x2^2 + 9352 x2^4 - 2310 x2^6 + 79 x2^8) α^3 μ^2 -

```

$$\begin{aligned}
& 75\,675\,600 (284 - 22 x^2 - 108 x^4 + 21 x^6) \alpha^2 \mu^3 + \\
& 1\,816\,214\,400 (-33 + 15 x^2 + x^4) \alpha \mu^4 + 27\,243\,216\,000 (-2 + x^2) \mu^5 + \\
& 7280 \text{Tf}^2 \mu (495 x^2 \alpha^2 (\alpha + 2 \mu) (\alpha + 3 \mu) ((2360 - 1036 x^2 - 252 x^4 + 117 x^6) \\
& \quad \alpha^2 + 672 (27 - 15 x^2 + x^4) \alpha \mu - 15\,120 (-2 + x^2) \mu^2) + \\
& x1 ((-208\,462 - 165 x^2 (3308 - 1510 x^2 - 252 x^4 + 117 x^6)) \alpha^6 - \\
& \quad 30 (126\,059 + 33 x^2 (7426 - 3737 x^2 - 140 x^4 + 117 x^6)) \alpha^5 \mu - \\
& \quad 99 (297\,328 + 15 x^2 (24\,312 - 12\,684 x^2 + 196 x^4 + 117 x^6)) \alpha^4 \mu^2 - \\
& \quad 23\,760 (5209 + 42 x^2 (77 - 40 x^2 + x^4)) \alpha^3 \mu^3 + 1\,995\,840 \\
& \quad (-146 + 15 x^2 (-2 + x^2)) \alpha^2 \mu^4 - 359\,251\,200 \alpha \mu^5 - 179\,625\,600 \mu^6) - \\
& 538\,137\,600 \alpha \mu^2 (\alpha + 3 \mu)^3 (60 \alpha \mu (x1 (47 + 6 (-6 + x1) x1) \mu + 6 (8 - 3 x1) \nu) - \\
& \quad 720 \mu^2 ((-4 + 3 x1) \mu + 2 (-2 + x1) \nu) + \\
& \quad \alpha^2 ((x1 (1627 + 120 (-7 + x1) x1) + 60 x1 x^2 - 30 (-2 + x1) x^4 - 120 (6 + x^2)) \\
& \quad \mu + 720 \nu - 218 x1 \nu)) + 4\,804\,800 \text{Tf} \mu (\alpha + 3 \mu)^2 \\
& (-20\,160 \alpha \mu^3 ((-5 + 3 x1 (-8 + 3 x1)) \mu - 5 \nu) + 60\,480 \mu^4 (\mu + \nu) - 168 \alpha^2 \mu^2 \\
& \quad ((-362 + 1080 (-3 + x1) x1 + 30 x^2 - 15 x^4) \mu - 362 \nu + 30 x^2 (-2 + x^2) \nu) + \\
& \quad \alpha^4 (3 (8 (56 + (942 - 247 x1) x1) - 2 (1363 + 280 (-5 + x1) x1) x^2 + 7 \\
& \quad (143 + 40 (-5 + x1) x1) x^4 + 308 x^6 - 50 x^8) \mu + \\
& \quad 2 (672 + 1354 x^2 - 917 x^4 + 196 x^6 - 27 x^8) \nu) + 84 \alpha^3 \mu \\
& \quad ((8 (23 + 292 x1 - 86 x1^2) - 3 (69 + 20 (-4 + x1) x1) x^2 + 15 (5 + 2 (-4 + x1) x1) \\
& \quad x^4 + 19 x^6) \mu + 2 (92 + 5 x^2 (13 - 8 x^2 + x^4) \nu)) , \\
& 1 / (7\,884\,800 (\alpha + 3 \mu)^6) \times 3 \text{Tf}^4 x2 \alpha^2 \mu (2640 \text{Tf} (-1 + x^2) \alpha \mu (\alpha + 3 \mu) \\
& \quad ((1038 - 491 x1 - 15 (-2 + x1) x^2 + 6 (-2 + x1) x^4) \alpha^2 - \\
& \quad 840 (-7 + 4 x1) \alpha \mu + 2520 (3 - 2 x1) \mu^2) + \\
& \text{Tf}^2 (-1 + x2) (1 + x2) ((-100\,105 + 14\,383 x^2 - 1259 x^4 + 853 x^6 - 192 x^8) \alpha^4 + \\
& \quad 1320 (-791 + 90 x^2 + 6 x^4) \alpha^3 \mu + 3960 (-1096 + 3 x^2 (36 + x^2)) \alpha^2 \mu^2 + \\
& \quad 166\,320 (-47 + 3 x^2) \alpha \mu^3 - 4\,989\,600 \mu^4) - 36\,960 (-1 + x^2) \alpha \mu (\alpha + 3 \mu)^2 \\
& \quad ((53 + 60 (-4 + x1) x1 + 93 x^2) \alpha \mu + 16 (-7 + 3 x^2) \alpha \nu - 60 \mu (9 \mu + 5 \nu))) \};
\end{aligned}$$

(*_ 8th order _*)

$$\begin{aligned}
\varphi_8 &= 1 / (7\,175\,168\,000 (\alpha + 3 \mu)^7) \times 3 \text{Tf}^4 \alpha \mu \\
& (-\text{Tf}^3 x1 (73\,047\,019 \alpha^6 + 1\,244\,789\,910 \alpha^5 \mu + 8\,706\,387\,690 \alpha^4 \mu^2 + 32\,735\,102\,400 \\
& \quad \alpha^3 \mu^3 + 70\,151\,281\,200 \alpha^2 \mu^4 + 81\,729\,648\,000 \alpha \mu^5 + 40\,864\,824\,000 \mu^6) - \\
& 2\,402\,400 \text{Tf} \alpha \mu (\alpha + 3 \mu)^2 (-15\,120 \mu^3 (-4 \mu + 9 x1 \mu - 4 \nu + 5 x1 \nu) + 2520 \alpha \mu^2 \\
& \quad (3 (10 - 6 x^2 + x1 (-8 - 6 x1 + x1^2 + 4 x^2)) \mu + (30 - 34 x1 - 9 x^2 + 6 x1 x^2) \\
& \quad \nu) + 42 \alpha^2 \mu ((724 - 840 x^2 + x1 (647 + 120 (-7 + x1) x1 + 480 x^2)) \mu + \\
& \quad (724 - 738 x1 + 30 (-12 + 7 x1) x^2 + 15 (-2 + x1) x^4) \nu) + \\
& \quad \alpha^3 (2 (1932 + 4 x1 (1448 + 105 (-8 + x1) x1) + 21 (-154 + 73 x1) x^2 - 105 \\
& \quad (-2 + x1) x^4 + 63 (-2 + x1) x^6) \mu + 3 (1288 - 1195 x1 + 14 \\
& \quad (-47 + 24 x1) x^2 + 35 (-7 + 3 x1) x^4 - 14 (-2 + x1) x^6) \nu) - \\
& 15\,015 \text{Tf}^2 (\alpha + 3 \mu) (-1\,814\,400 \mu^5 (\mu + \nu) + 302\,400 \alpha \mu^4 \\
& \quad ((-11 + 6 x1 (-8 + 3 x1) + 6 x^2) \mu + (-11 + 3 x^2) \nu) + \\
& \quad 3 \alpha^4 \mu (16 (-3311 + 10 x1 (-6116 + 1623 x1) + 11\,330 x^2 - 5600 x^4 + 1820 x^6) \mu + \\
& \quad (-52\,976 + 12\,760 x^2 + 18\,620 x^4 - 980 x^6 - 585 x^8) \nu) + \\
& \quad \alpha^5 (4 (-3096 + 40 x1 (-1677 + 398 x1) + 14\,628 x^2 - 8830 x^4 + 2828 \\
& \quad x^6 + 255 x^8 - 56 x^{10}) \mu + (-12\,384 + 4276 x^2 + 2810 \\
& \quad x^4 + 2156 x^6 - 1155 x^8 + 128 x^{10}) \nu) + 10\,080 \alpha^2 \mu^3 \\
& \quad ((-239 + 720 (-3 + x1) x1 + 300 x^2 - 30 x^4) \mu + (-239 + 15 x^2 (6 + x^2)) \nu) + \\
& \quad 240 \alpha^3 \mu^2 ((-3617 + 42 x1 (-1196 + 355 x1) + 42 x^2 (190 - 55 x^2 + 13 x^4)) \mu + \\
& \quad (-3617 + 21 x^2 (58 + 35 x^2 - 3 x^4)) \nu) - 504\,504\,000 \alpha \mu (\alpha + 3 \mu)^3 \\
& \quad (48 \mu^2 (\mu + \nu) (3 \mu + 2 \nu) + 4 \alpha \mu (-2 (19 + 18 (-4 + x1) x1 + 27 x^2) \mu^2 -
\end{aligned}$$

$$\begin{aligned} & 3 (5 + 6 (-4 + x_1) x_1 + 19 x_2^2) \mu \nu + (23 - 15 x_2^2) \nu^2) + \\ & \alpha^2 (2 (-4 (14 + x_1 (-32 + 7 x_1)) + 12 (-1 + (-4 + x_1) x_1) x_2^2 + 33 x_2^4) \mu^2 + \\ & (-90 + 4 (32 - 7 x_1) x_1 + 4 (-19 + 3 (-4 + x_1) x_1) x_2^2 + 65 x_2^4) \mu \nu + \\ & 2 (11 + 8 x_2^2 (-2 + x_2^2)) \nu^2)) \end{aligned}$$

u8 = {1 / (229 605 376 000 (α + 3 μ)^8) × 3 Tf^5 α μ

$$\begin{aligned} & (Tf^3 x1 (1471 678 450 \alpha^7 + 27 902 482 029 \alpha^6 \mu + 228 263 981 400 \alpha^5 \mu^2 + \\ & 1 040 579 499 960 \alpha^4 \mu^3 + 2 862 353 894 400 \alpha^3 \mu^4 + \\ & 4 789 357 372 800 \alpha^2 \mu^5 + 4 576 860 288 000 \alpha \mu^6 + 1 961 511 552 000 \mu^7) + \\ & 21 840 Tf^2 (\alpha + 3 \mu) (\alpha^6 ((-204 336 + x1 (-5 148 717 + 1 565 609 x1)) \mu - 204 336 \nu) + \\ & 33 \alpha^5 \mu ((-91 848 + 7 x1 (-288 212 + 98 013 x1)) \mu - 91 848 \nu) + \\ & 792 \alpha^4 \mu^2 ((-24 707 + 30 x1 (-14 798 + 5619 x1)) \mu - 24 707 \nu) + \\ & 15 840 \alpha^3 \mu^3 ((-4318 + 21 x1 (-2848 + 1199 x1)) \mu - 4318 \nu) + \\ & 1 330 560 \alpha^2 \mu^4 ((-101 + 75 x1 (-13 + 6 x1)) \mu - 101 \nu) + 19 958 400 \alpha \mu^5 \\ & ((-7 + 18 (-2 + x1) x1) \mu - 7 \nu) - 59 875 200 \mu^6 (\mu + \nu)) + 960 960 Tf \alpha \mu (\alpha + 3 \mu)^2 \\ & (\alpha^4 (3 (104 160 + x1 (654 741 + 40 x1 (-10 531 + 1574 x1))) \mu + 4 (78 120 - 52 469 x1) \nu) + \\ & 2520 \alpha^2 \mu^2 ((4696 + 3 x1 (3887 + 360 x1 (-11 + 2 x1))) \mu + 8 (587 - 484 x1) \nu) + \\ & 360 \alpha^3 \mu ((8932 + x1 (38 776 + 7 x1 (-4266 + 703 x1))) \mu + 8932 \nu - 6504 x1 \nu) + \\ & 302 400 \alpha \mu^3 ((62 + x1 (29 + 18 (-5 + x1) x1)) \mu + 62 \nu - 60 x1 \nu) - \\ & 1 814 400 \mu^4 (-6 \mu + 11 x1 \mu - 6 \nu + 7 x1 \nu)) + 1 614 412 800 \alpha \mu (\alpha + 3 \mu)^3 \\ & (360 \mu^3 (\mu + \nu) (7 \mu + 5 \nu) + 30 \alpha^2 \mu ((-229 + 4 (-4 + x1) x1 (-10 + 3 (-4 + x1) x1)) \\ & \mu^2 - 6 (63 + 2 x1 (-32 + 7 x1)) \mu \nu + 43 \nu^2) + \\ & 60 \alpha \mu^2 ((-179 + 6 (60 - 17 x1) x1) \mu^2 - 15 (9 + 4 (-4 + x1) x1) \mu \nu + 44 \nu^2) + \\ & \alpha^3 ((-435 + x1 (-2106 + 5 x1 (577 + 24 (-9 + x1) x1))) \mu^2 + \\ & (-3105 + 2388 x1 - 466 x1^2) \mu \nu + 210 \nu^2))) , \\ & 1 / (229 605 376 000 (\alpha + 3 \mu)^8) \times 3 Tf^5 x2 \alpha^2 \mu (-21 840 Tf^2 (-1 + x2^2) \alpha \mu \\ & (\alpha + 3 \mu) ((-913 251 + 671 868 x1 + (-398 099 + 181 972 x1) x2^2 + (7669 - 4676 x1) x2^4 + \\ & (14 137 - 5996 x1) x2^6 + 384 (-2 + x1) x2^8) \alpha^4 - 66 (177 194 - 135 941 x1 + \\ & 3 (18 078 - 9167 x1) x2^2 + 15 (-226 + 121 x1) x2^4 + 195 (-2 + x1) x2^6) \alpha^3 \mu - \\ & 23 760 (2474 - 1984 x1 + 402 x2^2 - 227 x1 x2^2 + (-18 + 11 x1) x2^4) \alpha^2 \mu^2 + \\ & 332 640 (-384 + 325 x1 + 3 (-8 + 5 x1) x2^2) \alpha \mu^3 + 9979 200 (-10 + 9 x1) \mu^4) + \\ & Tf^3 (-1 + x2) (1 + x2) ((-281 450 006 - 87 708 466 x2^2 - 49 039 108 x2^4 + \\ & 12 155 168 x2^6 + 4 991 583 x2^8 - 1 542 399 x2^{10} + 108 096 x2^{12}) \alpha^6 - \\ & 3 (1 826 244 241 + 465 419 321 x2^2 + 217 635 785 x2^4 - 80 639 335 x2^6 + \\ & 942 165 x2^8 + 841 155 x2^{10}) \alpha^5 \mu - \\ & 360 360 (112 219 + 31 279 x2^2 + 5119 x2^4 - 2237 x2^6 + 78 x2^8) \alpha^4 \mu^2 + \\ & 1 081 080 (-150 747 - 39 587 x2^2 - 695 x2^4 + 685 x2^6) \alpha^3 \mu^3 + \\ & 259 459 200 (-1513 + 6 x2^2 (-48 + x2^2)) \alpha^2 \mu^4 - 49 037 788 800 (11 + x2^2) \alpha \mu^5 - \\ & 326 918 592 000 \mu^6) - 1 614 412 800 (-1 + x2^2) \alpha^2 \mu^2 \\ & (\alpha + 3 \mu)^3 ((1266 - 594 x2^2 + x1 (-113 + 60 (-6 + x1) x1 + 297 x2^2)) \alpha \mu + \\ & 4 (243 - 109 x1 + 36 (-2 + x1) x2^2) \alpha \nu - 120 \mu ((-36 + 19 x1) \mu + 10 (-2 + x1) \nu)) - \\ & 960 960 Tf (-1 + x2^2) \alpha \mu (\alpha + 3 \mu)^2 (-302 400 \mu^3 (16 \mu + 9 \nu) + \\ & 2520 \alpha \mu^2 (5 (-107 + 12 x1 (-32 + 9 x1) + 51 x2^2) \mu + 12 (-91 + 9 x2^2) \nu) + \\ & 60 \alpha^2 \mu ((2 (9979 + 7119 (-4 + x1) x1) + 3 (2665 + 126 (-4 + x1) x1) x2^2 + 225 x2^4) \\ & \mu + 3 (-4866 + 706 x2^2 + 125 x2^4) \nu) + \\ & \alpha^3 (3 (114 539 + 30 529 x2^2 + 10 (8 x1 (-2319 + 508 x1) + 24 x1 (-59 + 13 x1) \\ & x2^2 + 3 (1 - 8 (-4 + x1) x1) x2^4 + 61 x2^6)) \mu + \\ & 2 (-42 932 + 1148 x2^2 + 6125 x2^4 - 925 x2^6) \nu))) ; \end{aligned}$$

```
(*Verification of equations 1-8*)
```

```
(*=====*)
```

```
(*Equation 1*)
```

```
eq1 = Grad[φ2, {x1, x2}] - μ Laplacian[u0, {x1, x2}] // FullSimplify;
test1 = (eq1 == {0, 0});
```

```
(*Equation 2*)
```

```
eq2 = Div[u2, {x1, x2}] + Grad[φ2, {x1, x2}].u0 // FullSimplify;
test2 = (eq2 == 0);
```

```
(*Equation 3*)
```

```
eq3 = Grad[φ4, {x1, x2}] -
      μ Laplacian[u2, {x1, x2}] - (μ + ν) Grad[Div[u2, {x1, x2}], {x1, x2}] +
      Grad[u0, {x1, x2}].u2 + Grad[u2, {x1, x2}].u0 // FullSimplify;
test3 = (eq3 == {0, 0});
```

```
(*Equation 4*)
```

```
eq4 = Div[u4, {x1, x2}] + Grad[φ2, {x1, x2}].u2 +
      Grad[φ4, {x1, x2}].u0 + φ2 Div[u2, {x1, x2}] // FullSimplify;
test4 = (eq4 == 0);
```

```
(*Equation 5*)
```

```
eq5 = Grad[φ6, {x1, x2}] -
      μ Laplacian[u4, {x1, x2}] - (μ + ν) Grad[Div[u4, {x1, x2}], {x1, x2}] +
      Grad[u0, {x1, x2}].u4 + Grad[u2, {x1, x2}].u2 + Grad[u4, {x1, x2}].u0 +
      φ2 Grad[u0, {x1, x2}].u2 + φ2 Grad[u2, {x1, x2}].u0 // FullSimplify;
test5 = (eq5 == {0, 0});
```

```
(*Equation 6*)
```

```
eq6 = Div[u6, {x1, x2}] + Grad[φ2, {x1, x2}].u4 + Grad[φ4, {x1, x2}].u2 +
      Grad[φ6, {x1, x2}].u0 + φ4 Div[u2, {x1, x2}] + φ2 Div[u4, {x1, x2}] // FullSimplify;
test6 = (eq6 == 0);
```

```
(*Equation 7-momentum for φ8*)
```

```
eq7 = Grad[φ8, {x1, x2}] - μ Laplacian[u6, {x1, x2}] - (μ + ν)
      Grad[Div[u6, {x1, x2}], {x1, x2}] + (Grad[u6, {x1, x2}].u0 + Grad[u0, {x1, x2}].u6) +
      (Grad[u4, {x1, x2}].u2 + Grad[u2, {x1, x2}].u4) +
      φ2 (Grad[u4, {x1, x2}].u0 + Grad[u2, {x1, x2}].u2 + Grad[u0, {x1, x2}].u4) +
      φ4 (Grad[u2, {x1, x2}].u0 + Grad[u0, {x1, x2}].u2) // FullSimplify;
test7 = (eq7 == {0, 0});
```

```
(*Equation 8-continuity for u8*)
```

```
eq8 = Div[u8, {x1, x2}] + Grad[φ2, {x1, x2}].u6 + Grad[φ4, {x1, x2}].u4 +
      Grad[φ6, {x1, x2}].u2 + Grad[φ8, {x1, x2}].u0 + φ6 Div[u2, {x1, x2}] +
      φ4 Div[u4, {x1, x2}] + φ2 Div[u6, {x1, x2}] // FullSimplify;
test8 = (eq8 == 0);
```

```
(*_ Boundary checks:v=0 at x2=±1 -*)
```

```
b21 = {FullSimplify[u2[[2]] /. x2 → 1] == 0, FullSimplify[u2[[2]] /. x2 → -1] == 0};
```

```

b22 = {FullSimplify[D[u2[[1]], x2] /. x2 → 1] == 0,
  FullSimplify[D[u2[[1]], x2] /. x2 → -1] == 0};
b41 = {FullSimplify[u4[[2]] /. x2 → 1] == 0, FullSimplify[u4[[2]] /. x2 → -1] == 0};
b42 = {FullSimplify[D[u4[[1]], x2] /. x2 → 1] == 0,
  FullSimplify[D[u4[[1]], x2] /. x2 → -1] == 0};
b61 = {FullSimplify[u6[[2]] /. x2 → 1] == 0, FullSimplify[u6[[2]] /. x2 → -1] == 0};
b62 = {FullSimplify[D[u6[[1]], x2] /. x2 → 1] == 0,
  FullSimplify[D[u6[[1]], x2] /. x2 → -1] == 0};
b81 = {FullSimplify[u8[[2]] /. x2 → 1] == 0, FullSimplify[u8[[2]] /. x2 → -1] == 0};
b82 = {FullSimplify[D[u8[[1]], x2] /. x2 → 1] == 0,
  FullSimplify[D[u8[[1]], x2] /. x2 → -1] == 0};
allBdry = And @@ Flatten[{b21, b22, b41, b42, b61, b62, b81, b82}];

```

```

(*=====*)
(*Output section*)
(*=====*)

```

```

Print["====="];
Print["Verification up to 8th order (x1,x2)"];
Print["====="];
Print["eq1: ", test1];
Print["eq2: ", test2];
Print["eq3: ", test3];
Print["eq4: ", test4];
Print["eq5: ", test5];
Print["eq6: ", test6];
Print["eq7: ", test7];
Print["eq8: ", test8];
Print["All main equations passed: ",
  test1 && test2 && test3 && test4 && test5 && test6 && test7 && test8];

```

```

Print["-----"];
Print["Boundary conditions (v = 0 at x2 = ±1):"];
Print[" u2: ", b21];
Print[" u4: ", b41];
Print[" u6: ", b61];
Print[" u8: ", b81];
Print["Boundary conditions (∂u/∂x2 = 0 at x2 = ±1):"];
Print[" u2: ", b22];
Print[" u4: ", b42];
Print[" u6: ", b62];
Print[" u8: ", b82];
Print["All boundary tests passed: ", allBdry];

```

```

Print["-----"];
Print["Overall status: ",
  test1 && test2 && test3 && test4 && test5 && test6 && test7 && test8 && allBdry];
Print["====="];

```

Total time: 0.280822 s

=====

Verification up to 8th order (x1,x2)

=====

eq1: True

eq2: True

eq3: True

eq4: True

eq5: True

eq6: True

eq7: True

eq8: True

All main equations passed: True

Boundary conditions ($v = 0$ at $x_2 = \pm 1$):

u2: {True, True}

u4: {True, True}

u6: {True, True}

u8: {True, True}

Boundary conditions ($\partial u / \partial x_2 = 0$ at $x_2 = \pm 1$):

u2: {True, True}

u4: {True, True}

u6: {True, True}

u8: {True, True}

All boundary tests passed: True

Overall status: True

=====

In[63]:=